Reg. No.										

MANIPAL INSTITUTE OF TECHNOLOGY

MANIPAL

(A constituent unit of MAHE, Manipal)

FIFTH SEMESTER BTECH. (E & C) DEGREE END SEMESTER EXAMINATION
DECEMBER 2020/JANUARY 2021
SUBJECT: ELECTRONIC PRODUCT DESIGN AND PACKAGING (ECE - 4303)

TIME: 3 HOURS **MAX. MARKS: 50**

Instructions to candidates

- Answer **ALL** questions.
- Missing data may be suitably assumed.

1A. Discuss the national policy for electronics product manufacturing with flow chart. Give its objectives with the components of the product planning.

1B. Explain reverse engineering and redesign methodology with necessary flow chart.

1C. Draw the circuit diagram for regulated power supply operating on 230V, 50HZ AC signal giving an output voltage of ± 12V and output current of 1A.The supply voltage variation can be taken as $\pm 15\%$ .The load regulation should be more than 0.5% and the output ripple is less than 0.2%. Calculate the peak value of the secondary voltage and turns ratio of the transformer.

(4+3+3)

2A. Discuss the need of heat sink in Electronic circuits. Explain The construction of any one type of heat sink with neat diagram and give its features.

2B. Discuss Miller indices with its different plane with illustration.

2C. A Transistor with Vce=20V and Ic= 1A has a 1°C/W junction to a case thermal resistance. If the value of $\theta_{cs} = 0.4$°C. Calculate the thermal resistance for the heat sink that will keep the maximum junction temperature at 90°C when the ambient temperature is 25°C.

(4+3+3)

3A. Discuss various types of common IC packages. Explain wafer bumping technology and give its advantages.

3B. Discuss PCB manufacturing technics with flow chart. Explain the need of plating in Printed Circuit Boards.

3C. In a multilayer PCB signal and ground plane is separated by 0.15 inch, common area of two planes is 5.25 inch^2. Find the parasitic capacitance for relative permittivity of substrate with $\varepsilon_r = 2.5$.

(4+3+3)

4A. Discuss various types of noises in the Electronic circuits. Draw the block diagram of the low noise amplifier and Explain its function.

4B. Discuss the reliability issues in the integrated circuits with its consequences.

4C. Discuss the cooling choices with its transfer curve in the Electronic circuits

(4+3+3)

5A. What are the various types of electromagnetic interference in the electronic circuits? Give its effects and explain how to neutralize them.

5B. With neat diagram explain the cross talk effects in the electronic circuits. Give the remedies to minimize the cross talks.

5C. Explain the working of switched mode power supply with its block diagram and input output wave forms.

(4+3+3)